

ASSIGNMENT : 4 – BASIC COMPUTER HARDWARE

GUIDED BY : VIPUL MISTRI

Q.1 Open your computer or laptop and take a clear photo or screenshot of the System Information or About page that shows your processor (CPU), RAM, and storage details.

ANS :

The screenshot displays the Windows Settings application, specifically the 'About' page. The left sidebar shows the 'Settings' menu with options like Home, Find a setting, System, Display, Sound, Notifications & actions, Focus assist, Power & sleep, Storage, Tablet, Multitasking, Projecting to this PC, Shared experiences, System Components, Clipboard, Remote Desktop, and Optional features. The main content area is titled 'About' and shows system information in a grid format. The grid includes sections for Installed RAM (16.0 GB), Processor (Intel(R) Core(TM) i7-9700F CPU @ 3.00GHz), Graphics Card (4 GB), and Storage (462 GB). Below these, there is a section for 'DESKTOP-AK25G64' with a 'Rename this PC' button. A 'Device Specifications' section is also present, listing details such as Device Name, Processor, Installed RAM, Graphics Card, Storage, Device ID, Product ID, System Type, and Pen and touch. At the bottom, there is a 'Frequently asked questions' section with two questions related to RAM and GPU performance. The taskbar at the bottom shows the Windows logo, a search bar, and several application icons. The system tray on the right indicates the time as 2:44 PM on 06-Oct-26 and shows network and volume icons.

Settings

Home

Find a setting

System

Display

Sound

Notifications & actions

Focus assist

Power & sleep

Storage

Tablet

Multitasking

Projecting to this PC

Shared experiences

System Components

Clipboard

Remote Desktop

Optional features

About

Installed RAM

16.0 GB

Speed: 2666 MHz

Processor

Intel(R) Core(TM) i7-9700F CPU @ 3.00GHz

3.00 GHz

Graphics Card

4 GB

NVIDIA GeForce GT 730

Storage

462 GB

262 GB of 462 GB used

DESKTOP-AK25G64

System Product Name

Rename this PC

Device Specifications

Copy

Device Name

DESKTOP-AK25G64

Processor

Intel(R) Core(TM) i7-9700F CPU @ 3.00GHz 3.00 GHz

Installed RAM

16.0 GB

Graphics Card

NVIDIA GeForce GT 730 (4 GB)

Storage

238 GB SSD 256GB, 224 GB SSD CT240BX500SSD1

Device ID

FCCCF6E4-BE34-46C9-874D-D502F949C6CF

Product ID

00331-10000-00001-AA965

System Type

64-bit operating system, x64-based processor

Pen and touch

No pen or touch input is available for this display

Frequently asked questions

How does having 16 GB or more of RAM impact my PC's performance? Can I run modern applications smoothly with this RAM capacity?

Is my GPU sufficient for high end gaming and video experience? How can having a dedicated GPU enhance my experience and productivity?

BitLocker settings

Device Manager

Remote desktop

System protection

Advanced system settings

Rename this PC (advanced)

Help from the web

Finding out how many cores my processor has

Checking multiple Languages support

Get help

Give feedback

RELIANCE +2.45%

2:44 PM

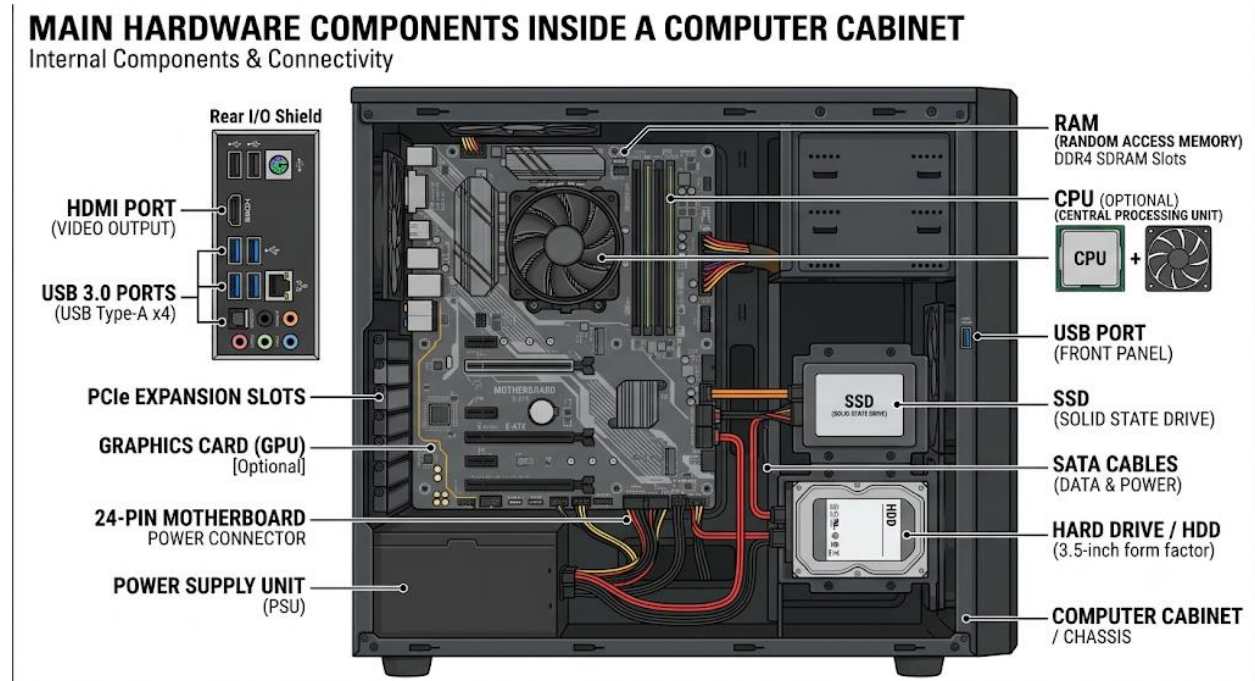
06-Oct-26

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Q.2 Draw and label a diagram (on paper or digitally) showing the main hardware components inside a computer cabinet, including CPU, RAM, motherboard, hard drive/SSD, power supply, and at least two ports (USB, HDMI, etc.).

ANS :



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Q.3 List the hardware components you would check or upgrade to make your laptop run faster for gaming or video editing, and explain in one line why each upgrade helps. Hint: Think about RAM, SSD, GPU, and cooling.

ANS : Here are the key hardware components to check or upgrade to boost laptop performance for gaming and video editing:

1. **RAM (Random Access Memory):** Increasing RAM capacity (e.g., upgrading to 16GB or 32GB) prevents system stuttering and allows smooth handling of heavy games, multi-track timeline editing, and background applications.
2. **SSD (Solid State Drive / NVMe):** Replacing an older HDD or adding a faster NVMe SSD drastically reduces game load times, speeds up boot times, and enables rapid scrubbing through high-resolution video files.
3. **GPU (Graphics Processing Unit):** Upgrading or leveraging a dedicated GPU (or using an eGPU enclosure) handles real-time 3D graphics rendering, frame generation in games, and hardware-accelerated video exporting.
4. **Thermal Cooling System (Thermal Paste & Cooling Pad):** Cleaning dust, reapplying high-performance thermal paste, or using an active cooling pad prevents thermal throttling, allowing your CPU and GPU to sustain maximum performance under heavy workloads.
5. **CPU (Central Processing Unit):** Though rarely upgradeable in laptops, optimizing or switching to a laptop with higher core counts speeds up complex physics processing in games and speeds up video rendering and encoding tasks.

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Q.4 Research the difference between SSD and HDD storage, then write a short comparison (3 points) explaining which one is better for a student who wants fast boot times and why.

ANS :

Here are 3 key differences between SSD and HDD storage:

1. **Speed & Performance:** SSDs (Solid State Drives) use flash memory and have no moving parts, delivering read/write speeds that are 4 to 10 times faster than traditional HDDs (Hard Disk Drives), which rely on spinning magnetic platters.
2. **Durability & Reliability:** SSDs are far more physical-impact resistant than HDDs because they contain no mechanical components, making them much less likely to fail or lose data if a laptop gets bumped or dropped.
3. **Noise, Power, & Efficiency:** SSDs run completely silently, consume less power, and generate less heat than HDDs, helping extend laptop battery life.

Which is better for fast boot times?

An SSD is significantly better. Because an SSD loads operating system files almost instantaneously without waiting for physical disk platters to spin up, it can boot a computer in under 10–15 seconds compared to 45–60+ seconds on an HDD. For a student needing quick access to notes and projects between classes, an SSD offers a vastly faster and more responsive experience.